HWRS 561b Homework #5

- Assigned: Monday, 5 April 2027
- Due: Thursday, 15 April 2027 at 10:00 pm MST (tentative; upload answers in PDF or Jupyter Notebook to D2L)
- Instructor: Ty Ferré (tyferre@arizona.edu)
- Teaching Assistant: Nathan Strom (nstrom@arizona.edu)
- Semester: Spring 2027

1. (20 points) Qualitative behavior of transient unsaturated flow. Consider two soils: a sandy loam and a silt.

   - (a) (8 points) Constant-potential infiltration. The surface pressure head is suddenly changed from an initially uniform $h_i = -40$ cm to $h_0 = -8$ cm. Explain why the wetting front is typically abrupt and narrow in sandy loam but broader and more gradual in silt. Also explain why the wetting front moves fastest at early times and why the infiltration rate decreases toward an almost constant value at later times.
   - (b) (6 points) Upward flow from a water table. In class, we showed that, at long times, silt can sustain a much larger upward evaporation-driven flux than sandy loam. Explain why this can happen even though coarse soils often have a larger saturated hydraulic conductivity.
   - (c) (6 points) Infiltration-evaporation cycles. Suppose the surface flux alternates every 12 h between infiltration and evaporation. Explain why pressure-head fluctuations attenuate with depth and why shorter-period forcing penetrates less deeply into the profile.

2. (20 points) Cell-centered finite difference solution of the transient 1D Richards equation. Let z be positive upward. Consider the mixed-form transient 1D Richards equation

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left[K(h)\frac{\partial h}{\partial z}\right] + \frac{\partial K(h)}{\partial z},$$

   on the domain $z \in [0, 1.0]$ m. The boundary conditions are

$$h(0,t) = 0 \text{ m}, \qquad h(1.0,t) = -1.0 \text{ m},$$

   and the initial condition is uniform:

$$h(z,0) = -1.0 \text{ m}.$$

   Use the following sand hydraulic properties with the van Genuchten-Mualem model:

$$\theta_r = 0.045, \qquad \theta_s = 0.43, \qquad \alpha = 0.145 \text{ cm}^{-1} = 14.5 \text{ m}^{-1}, \qquad n = 2.68,$$

$$m = 1 - \frac{1}{n} = 0.627, \qquad l = 0.5,$$

$$K_s = 712.8 \text{ cm/day} = 29.7 \text{ cm/h} = 8.25 \times 10^{-5} \text{ m/s}.$$

For the nonlinear system in part (b), use

$$\theta(h) = \theta_r + \frac{\theta_s - \theta_r}{\left(1 + |\alpha h|^n\right)^m}, \qquad h < 0,$$

with $\theta = \theta_s$ for $h \geq 0$. Also let $C(h) = d\theta/dh$. For conductivity, use

$$K(S_e) = K_s\, S_e^l \left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2, \qquad S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r}.$$

Discretize the domain into 4 uniform boxes ($\Delta z = 0.25$ m) using a cell-centered finite difference scheme, with cell-center heads h_1, h_2, h_3, h_4. Following the same spirit as Homework 4, approximate the boundary condition by treating h_1 and h_4 as boundary-representative cell values at all time levels, and take

$$h_1^{n+1} = h(0.125, t^{n+1}) = -0.125 \text{ m}, \qquad h_4^{n+1} = h(0.875, t^{n+1}) = -0.875 \text{ m}.$$

Thus, the unknown heads at the new time level are h_2^{n+1} and h_3^{n+1}. Use backward Euler (like we did in class) with one time step of $\Delta t = 1$ h. Evaluate the conductivity at each interior face by arithmetic averaging of the conductivities at the two adjacent cell centers.

For the frozen-coefficient calculation in part (c), use the old-time heads

$$h_1^n = -0.125 \text{ m}, \qquad h_2^n = -1.0 \text{ m}, \qquad h_3^n = -1.0 \text{ m}, \qquad h_4^n = -0.875 \text{ m},$$

For the initial Picard iterate, set

$$h_2^{n+1,0} = h_2^n = -1.0 \text{ m}, \qquad h_3^{n+1,0} = h_3^n = -1.0 \text{ m}.$$

Together with the boundary-representative values $h_1^{n+1} = -0.125$ m and $h_4^{n+1} = -0.875$ m, this gives the old-time water contents

$$\theta_1^n = 1.71 \times 10^{-1}, \qquad \theta_2^n = \theta_3^n = 4.93 \times 10^{-2}, \qquad \theta_4^n = 5.04 \times 10^{-2},$$

the frozen Picard water contents and cell capacities

$$\theta_2^{n+1,0} = \theta_3^{n+1,0} = 4.93 \times 10^{-2},$$

$$C_2^{n+1,0} = C_3^{n+1,0} = 7.23 \times 10^{-3} \text{ m}^{-1},$$

and the following frozen face conductivities:

$$K_{3/2}^{n+1,0} = 2.83 \times 10^{-7}, \qquad K_{5/2}^{n+1,0} = 2.04 \times 10^{-12}, \qquad K_{7/2}^{n+1,0} = 3.35 \times 10^{-12} \text{ m/s}.$$

No coding is required for this problem. Derive and solve the system by hand.

- (a) (8 points) Using cell-centered finite differences and backward Euler (like we did in class), derive the algebraic equation for each interior cell ($i = 2, 3$) marching from time step n to $n+1$, including the transient storage term written as $\left(\theta_i^{n+1} - \theta_i^n\right)/\Delta t$ and the gravity term written as $\left(K_{i+1/2} - K_{i-1/2}\right)/\Delta z$. Treat h_1^{n+1} and h_4^{n+1} as known from the boundary approximation above.
- (b) (6 points) Write the coupled nonlinear system in residual form, $\mathbf{F}(\mathbf{h}^{n+1}) = \mathbf{0}$, for $\mathbf{h}^{n+1} = (h_2^{n+1}, h_3^{n+1})^T$ after substituting the known old-time heads and the given values of h_1^{n+1} and h_4^{n+1}.
- (c) (6 points) Linearize the mixed-form storage term with one Picard step,

$$\theta_i^{n+1,1} \approx \theta_i^{n+1,0} + C_i^{n+1,0}\left(h_i^{n+1,1} - h_i^{n+1,0}\right),$$

using the frozen values of $\theta_i^{n+1,0}$, $C_i^{n+1,0}$, and $K_{i+1/2}^{n+1,0}$ given above. Assemble the resulting 4×4 linear system and solve for the first Picard iterate $h_2^{n+1,1}$ and $h_3^{n+1,1}$ ($h_1^{n+1,1}$ and $h_4^{n+1,1}$ are given from boundary conditions). Furthermore, explain in words (no need to actually calculate) how you would move forward to obtain the converged solution for time step $n+1$.

3. (20 points) Construct a 100 cm long vertical column in HYDRUS-1D. Fill it with the default sand, initially at field capacity. Apply an infiltration flux equal to $K_{sat}/10$ at the top boundary. Apply a physically natural bottom boundary condition and justify your choice.

   You can find hydraulic parameters for the default sand in HYDRUS (e.g., $K_{sat} = 0.00825$ cm/s). In HYDRUS, a negative top flux represents infiltration and a positive top flux represents evapotranspiration. For the hand calculation, determine the water content in the wetted region from the van Genuchten-Mualem functions for the default sand.

   - (a) (6 points) Under the piston-displacement assumption, determine the initial water content at field capacity, θ_i, from the HYDRUS default condition and the uniform water content behind the wetting front, θ_w, corresponding to the imposed infiltration flux $K_{sat}/10$. Explain briefly how you obtained both values.
   - (b) (8 points) Following (a), derive an expression for the wetting-front depth $z_f(t)$. Sketch the corresponding hand-calculated water-content profile at a range of times that space relatively evenly.
   - (c) (6 points) Run HYDRUS-1D for the elapsed time you computed in part (b). Compare the HYDRUS water-content profiles with your piston-displacement profiles. Discuss the main differences and explain why.

4. (20 points) Use HYDRUS-1D to construct a model of a 1 m long sand profile with the default sand parameters. The profile is initially fully saturated with a constant pressure head of 0 cm. Impose free drainage at the bottom boundary. At time $t = 0$, apply a zero-flux top boundary condition and simulate 10,000 s. Include an observation node at 50 cm depth.

   - (a) (8 points) Plot the water content at 50 cm depth as a function of time. Then assume that the water content remains uniform throughout the domain during drainage. Based on mass balance, use the water-content time series at the observation node to estimate the bottom flux as a function of time. In addition, compute the flux using the same approach from Problem 5(c) in Homework 4. On the same axes, plot the bottom flux reported directly by HYDRUS and the two fluxes you computed. Comment on the differences between the three flux estimates.

- (b) (6 points) Assuming unit-gradient drainage, compute K as a function of the observed θ using: 1) the flux estimated from mass balance in part (a), and 2) the flux reported directly by HYDRUS. Plot the two inferred $K(\theta)$ relationships together with the correct $K(\theta)$ for the default sand used in the forward model. Comment on the differences among the three curves. Hint: You are not allowed to use any of the parameters you use in HYDRUS to do your calculations. Think of the HYDRUS calculation as an "experiment" that someone else did it for you, thus you should not know any of the parameters that went into the HYDRUS calculation. You should only compare to the true K vs. θ after you finish your calculations.
- (c) (6 points) First fix $K_s = 0.00825$ cm/s and $\theta_r = 0.045$ cm^3/cm^3. Fit the Mualem form of the van Genuchten relationship to the $K(\theta)$ measurements inferred from the HYDRUS bottom-flux data, and report your best-fit n value. Then plot paired measurements of pressure head and water content at the observation point and fit the van Genuchten retention model to estimate α and n. Compare your fitted parameter values with the forward-model values and comment on any differences.

5. (20 points) Use HYDRUS-1D to estimate van Genuchten parameters for the default sand by inverse modeling. Base this problem on the same 1 m drainage experiment from Problem 4: the profile is initially fully saturated with a constant pressure head of 0 cm, the top boundary is zero flux for $t > 0$, the bottom boundary is free drainage, the total simulation time is 10,000 s, and there are three observation nodes at 25 cm, 50 cm, and 75 cm depths. Treat the forward simulation with the default sand as the synthetic "measured" data set.

   Suggested approach for measurement noise: after exporting the synthetic water content data, you may optionally add Gaussian noise to mimic measurement error using

$$\theta_{obs} = \theta_{true} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_\theta^2).$$

   A reasonable choice is $\sigma_\theta = 0.003$ to 0.005 cm^3/cm^3. If you add noise, use a fixed random seed and keep the same noisy data set for all inverse runs so the comparisons are fair.

   - (a) (5 points) Run the forward model with the default sand and export the water content time series at 25 cm, 50 cm, and 75 cm depths. State clearly whether you used the raw synthetic data or a Gaussian-noise-perturbed version of the data for the inverse modeling, and report the noise level if you added noise.
   - (b) (5 points) In HYDRUS-1D inverse mode, estimate α and n while holding K_s, θ_r, and θ_s fixed at their default sand values. For this run, only use the water content data at 50 cm depth. Report the fitted parameters and compare them with the true forward-model values.
   - (c) (5 points) Repeat the inverse modeling using the water content data at all three depths. Compare the estimated parameters from the two inverse runs and explain whether the multi-depth data set constrains the parameters better.
   - (d) (5 points) Add 5% Gaussian noise to the water content data and repeat the inverse modeling in part (c). In other words, perturb the synthetic data using

$$\theta_{obs} = \theta_{true} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma_\theta^2), \qquad \sigma_\theta = 0.05\,\theta_{true}.$$

Use a fixed random seed. Compare the estimated parameters from the noisy-data inversion with those from part (c), and comment on how the added measurement errors affected the parameter estimates and the overall fit.